

Project 2 report

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In this project, we enhanced the CUDA implementation of the Spatial Distance Histogram (SDH) computation in Project 1 by applying optimized memory management, efficient threaded workload allocation, and memory contention reduction through shared memory utilization and atomic operations. While Project 1 used a global memory approach that resulted in high contention and inefficient memory accesses, Project 2 introduced advanced optimizations including shared memory usage, merged memory accesses, and atomic operation minimization to achieve a more efficient CUDA kernel.

A key optimization in Project 2 is output privatization, which reduces atomic contention by implementing a private histogram for each block in shared memory. Instead of performing frequent nuclear operations on the global histogram, each thread block first computes the local histogram and then merges the results into global memory, minimizing global memory atomic updates and significantly increasing execution speed.

Another significant improvement is memory access optimization, as Project 1 exhibits high global memory latency. To address this issue, we allocated shared memory for each thread block, reduced frequent accesses to global memory, and structured the data to ensure consolidated memory accesses, minimizing unaligned memory transactions and improving overall efficiency.

In addition, workload allocation and thread optimization were improved by dynamically allocating the computational scope of threads to prevent redundant computations and ensure an even distribution of workloads among threads. We also experimented with different thread block sizes (256 and 512) to determine the optimal configuration and ultimately found that 256 threads per block provided a better balance of computation and memory overhead than 512 threads, which introduced higher shared memory contention.

Project 2 for `/apps/GPU_course/runScript.sh proj2.cu 10000 500 128`

```

srund: route/default: init: route default plugin loaded
srund: launch/slurm: _task_start: Node GPU13, 1 tasks started

***** Total Running Time of CPU = 1.06956 sec *****
00: 2076 | 01: 14212 | 02: 37870 | 03: 70863 | 04: 113190 |
05: 162616 | 06: 219792 | 07: 281373 | 08: 348231 | 09: 418908 |
10: 492542 | 11: 568919 | 12: 643293 | 13: 723787 | 14: 799933 |
15: 878146 | 16: 953657 | 17: 1025635 | 18: 1097551 | 19: 1161963 |
20: 1225131 | 21: 1285543 | 22: 1340874 | 23: 1389657 | 24: 1435607 |
25: 1471150 | 26: 1505326 | 27: 1534652 | 28: 1554499 | 29: 1566213 |
30: 1574002 | 31: 1572648 | 32: 1562633 | 33: 1550305 | 34: 1527753 |
35: 1500146 | 36: 1463004 | 37: 1419940 | 38: 1372128 | 39: 1313777 |
40: 1249134 | 41: 1177114 | 42: 1097835 | 43: 1012343 | 44: 920271 |
45: 823626 | 46: 723835 | 47: 632337 | 48: 548947 | 49: 472904 |
50: 405296 | 51: 344273 | 52: 289347 | 53: 240911 | 54: 197652 |
55: 160654 | 56: 129093 | 57: 101564 | 58: 78499 | 59: 60069 |
60: 44360 | 61: 32554 | 62: 23054 | 63: 16448 | 64: 11533 |
65: 7835 | 66: 5295 | 67: 3371 | 68: 2289 | 69: 1372 |
70: 804 | 71: 450 | 72: 222 | 73: 104 | 74: 34 |
75: 16 | 76: 8 | 77: 2 | 78: 0 | 79: 0 |
T:49995000

***** Total Running Time of GPU Kernel = 21.61002 ms *****
00: 2076 | 01: 14212 | 02: 37870 | 03: 70863 | 04: 113190 |
05: 162616 | 06: 219792 | 07: 281373 | 08: 348231 | 09: 418908 |
10: 492542 | 11: 568919 | 12: 643293 | 13: 723787 | 14: 799933 |
15: 878146 | 16: 953657 | 17: 1025635 | 18: 1097551 | 19: 1161963 |
20: 1225131 | 21: 1285543 | 22: 1340874 | 23: 1389657 | 24: 1435607 |
25: 1471150 | 26: 1505326 | 27: 1534652 | 28: 1554499 | 29: 1566213 |
30: 1574002 | 31: 1572648 | 32: 1562633 | 33: 1550305 | 34: 1527753 |
35: 1500146 | 36: 1463004 | 37: 1419940 | 38: 1372128 | 39: 1313777 |
40: 1249134 | 41: 1177114 | 42: 1097835 | 43: 1012343 | 44: 920271 |
45: 823626 | 46: 723835 | 47: 632337 | 48: 548947 | 49: 472904 |
50: 405296 | 51: 344273 | 52: 289347 | 53: 240911 | 54: 197652 |
55: 160654 | 56: 129093 | 57: 101564 | 58: 78499 | 59: 60069 |
60: 44360 | 61: 32554 | 62: 23054 | 63: 16448 | 64: 11533 |
65: 7835 | 66: 5295 | 67: 3371 | 68: 2289 | 69: 1372 |
70: 804 | 71: 450 | 72: 222 | 73: 104 | 74: 34 |
75: 16 | 76: 8 | 77: 2 | 78: 0 | 79: 0 |
T:49995000

Difference between CPU and GPU histograms:
00: 0 | 01: 0 | 02: 0 | 03: 0 | 04: 0 |
05: 0 | 06: 0 | 07: 0 | 08: 0 | 09: 0 |
10: 0 | 11: 0 | 12: 0 | 13: 0 | 14: 0 |
15: 0 | 16: 0 | 17: 0 | 18: 0 | 19: 0 |
20: 0 | 21: 0 | 22: 0 | 23: 0 | 24: 0 |
25: 0 | 26: 0 | 27: 0 | 28: 0 | 29: 0 |
30: 0 | 31: 0 | 32: 0 | 33: 0 | 34: 0 |

```

Project 2 for /apps/GPU_course/runScript.sh proj2.cu 10000 500 256

```

srund: end of defined options
srund: jobid 228781: nodes(1):'GPU13', cpu counts: 16(x1)
srund: launch/slurm: launch_p_step_launch: CpuBindType=(null type)
srund: launching StepId=228781.0 on host GPU13, 1 tasks: 0
srund: route/default: init: route default plugin loaded
srund: launch/slurm: _task_start: Node GPU13, 1 tasks started

***** Total Running Time of CPU = 1.06919 sec *****
00: 2076 | 01: 14212 | 02: 37870 | 03: 70863 | 04: 113190 |
05: 162616 | 06: 219792 | 07: 281373 | 08: 348231 | 09: 418908 |
10: 492542 | 11: 568919 | 12: 643293 | 13: 723787 | 14: 799933 |
15: 878146 | 16: 953657 | 17: 1025635 | 18: 1097551 | 19: 1161963 |
20: 1225131 | 21: 1285543 | 22: 1340874 | 23: 1389657 | 24: 1435607 |
25: 1471150 | 26: 1505326 | 27: 1534652 | 28: 1554499 | 29: 1566213 |
30: 1574002 | 31: 1572648 | 32: 1562633 | 33: 1550305 | 34: 1527753 |
35: 1500146 | 36: 1463004 | 37: 1419940 | 38: 1372128 | 39: 1313777 |
40: 1249134 | 41: 1177114 | 42: 1097835 | 43: 1012343 | 44: 920271 |
45: 823626 | 46: 723835 | 47: 632337 | 48: 548947 | 49: 472904 |
50: 405296 | 51: 344273 | 52: 289347 | 53: 240911 | 54: 197652 |
55: 160654 | 56: 129093 | 57: 101564 | 58: 78499 | 59: 60069 |
60: 44360 | 61: 32554 | 62: 23054 | 63: 16448 | 64: 11533 |
65: 7835 | 66: 5295 | 67: 3371 | 68: 2289 | 69: 1372 |
70: 804 | 71: 450 | 72: 222 | 73: 104 | 74: 34 |
75: 16 | 76: 8 | 77: 2 | 78: 0 | 79: 0 |
T:49995000

***** Total Running Time of GPU Kernel = 25.67629 ms *****
00: 2076 | 01: 14212 | 02: 37870 | 03: 70863 | 04: 113190 |
05: 162616 | 06: 219792 | 07: 281373 | 08: 348231 | 09: 418908 |
10: 492542 | 11: 568919 | 12: 643293 | 13: 723787 | 14: 799933 |
15: 878146 | 16: 953657 | 17: 1025635 | 18: 1097551 | 19: 1161963 |
20: 1225131 | 21: 1285543 | 22: 1340874 | 23: 1389657 | 24: 1435607 |
25: 1471150 | 26: 1505326 | 27: 1534652 | 28: 1554499 | 29: 1566213 |
30: 1574002 | 31: 1572648 | 32: 1562633 | 33: 1550305 | 34: 1527753 |
35: 1500146 | 36: 1463004 | 37: 1419940 | 38: 1372128 | 39: 1313777 |
40: 1249134 | 41: 1177114 | 42: 1097835 | 43: 1012343 | 44: 920271 |
45: 823626 | 46: 723835 | 47: 632337 | 48: 548947 | 49: 472904 |
50: 405296 | 51: 344273 | 52: 289347 | 53: 240911 | 54: 197652 |
55: 160654 | 56: 129093 | 57: 101564 | 58: 78499 | 59: 60069 |
60: 44360 | 61: 32554 | 62: 23054 | 63: 16448 | 64: 11533 |
65: 7835 | 66: 5295 | 67: 3371 | 68: 2289 | 69: 1372 |
70: 804 | 71: 450 | 72: 222 | 73: 104 | 74: 34 |
75: 16 | 76: 8 | 77: 2 | 78: 0 | 79: 0 |
T:49995000

Difference between CPU and GPU histograms:
00: 0 | 01: 0 | 02: 0 | 03: 0 | 04: 0 |
05: 0 | 06: 0 | 07: 0 | 08: 0 | 09: 0 |
10: 0 | 11: 0 | 12: 0 | 13: 0 | 14: 0 |
15: 0 | 16: 0 | 17: 0 | 18: 0 | 19: 0 |

```

Project 2 for /apps/GPU_course/runScript.sh proj2.cu 10000 500 512

```

srun: jobid 228783: nodes(1):'GPU13', cpu counts: 16(x1)
srun: launch/slurm: launch.p_step_launch: CpuBindType=(null type)
srun: launching StepId=228783.0 on host GPU13, 1 tasks: 0
srun: route/default: init: route default plugin loaded
srun: launch/slurm: _task_start: Node GPU13, 1 tasks started

***** Total Running Time of CPU = 1.06839 sec *****
00:      2076 | 01:      14212 | 02:      37870 | 03:      70863 | 04:      113190 |
05:     162616 | 06:     219792 | 07:     281373 | 08:     348231 | 09:     418908 |
10:     492542 | 11:     568919 | 12:     643293 | 13:     723787 | 14:     799933 |
15:     878146 | 16:     953657 | 17:    1025635 | 18:    1097551 | 19:    1161963 |
20:    1225131 | 21:    1285543 | 22:    1340874 | 23:    1389657 | 24:    1435607 |
25:    1471150 | 26:    1505326 | 27:    1534652 | 28:    1554499 | 29:    1566213 |
30:    1574002 | 31:    1572648 | 32:    1562633 | 33:    1550305 | 34:    1527753 |
35:    1500146 | 36:    1463004 | 37:    1419940 | 38:    1372128 | 39:    1313777 |
40:    1249134 | 41:    1177114 | 42:    1097835 | 43:    1012343 | 44:     920271 |
45:     823626 | 46:     723835 | 47:     632337 | 48:     548947 | 49:     472904 |
50:     405296 | 51:     344273 | 52:     289347 | 53:     240911 | 54:     197652 |
55:     160654 | 56:     129093 | 57:     101564 | 58:     78499  | 59:     60069  |
60:     44360  | 61:     32554  | 62:     23054  | 63:     16448  | 64:     11533  |
65:      7835  | 66:      5295  | 67:      3371  | 68:      2289  | 69:      1372  |
70:       804  | 71:       450  | 72:       222  | 73:       104  | 74:        34  |
75:        16  | 76:         8  | 77:         2  | 78:         0  | 79:         0  |
T:49995000

***** Total Running Time of GPU Kernel = 31.37904 ms *****
00:      2076 | 01:      14212 | 02:      37870 | 03:      70863 | 04:      113190 |
05:     162616 | 06:     219792 | 07:     281373 | 08:     348231 | 09:     418908 |
10:     492542 | 11:     568919 | 12:     643293 | 13:     723787 | 14:     799933 |
15:     878146 | 16:     953657 | 17:    1025635 | 18:    1097551 | 19:    1161963 |
20:    1225131 | 21:    1285543 | 22:    1340874 | 23:    1389657 | 24:    1435607 |
25:    1471150 | 26:    1505326 | 27:    1534652 | 28:    1554499 | 29:    1566213 |
30:    1574002 | 31:    1572648 | 32:    1562633 | 33:    1550305 | 34:    1527753 |
35:    1500146 | 36:    1463004 | 37:    1419940 | 38:    1372128 | 39:    1313777 |
40:    1249134 | 41:    1177114 | 42:    1097835 | 43:    1012343 | 44:     920271 |
45:     823626 | 46:     723835 | 47:     632337 | 48:     548947 | 49:     472904 |
50:     405296 | 51:     344273 | 52:     289347 | 53:     240911 | 54:     197652 |
55:     160654 | 56:     129093 | 57:     101564 | 58:     78499  | 59:     60069  |
60:     44360  | 61:     32554  | 62:     23054  | 63:     16448  | 64:     11533  |
65:      7835  | 66:      5295  | 67:      3371  | 68:      2289  | 69:      1372  |
70:       804  | 71:       450  | 72:       222  | 73:       104  | 74:        34  |
75:        16  | 76:         8  | 77:         2  | 78:         0  | 79:         0  |
T:49995000

Difference between CPU and GPU histograms:
00:      0 | 01:      0 | 02:      0 | 03:      0 | 04:      0 |
05:      0 | 06:      0 | 07:      0 | 08:      0 | 09:      0 |
10:      0 | 11:      0 | 12:      0 | 13:      0 | 14:      0 |
15:      0 | 16:      0 | 17:      0 | 18:      0 | 19:      0 |

```

Project 1 for apps/GPU_course/runScript.sh proj1 -feng12.cu 10000 500 128

```

srun: route/default: init: route default plugin loaded
srun: launch/slurm: _task_start: Node GPU15, 1 tasks started
Running time: 1.339542
00:      2076 |      14212 |      37870 |      70863 |      113190 |
05:      162616 |      219792 |      281373 |      348231 |      418908 |
10:      492542 |      568919 |      643293 |      723787 |      799933 |
15:      878146 |      953657 |      1025635 |      1097551 |      1161963 |
20:      1225131 |      1285543 |      1340874 |      1389657 |      1435607 |
25:      1471150 |      1505326 |      1534652 |      1554099 |      1566213 |
30:      1574002 |      1572648 |      1562633 |      1558305 |      1527753 |
35:      1580146 |      1463004 |      1419940 |      1372128 |      1313777 |
40:      1249134 |      1177114 |      1097835 |      1012343 |      928271 |
45:      823626 |      723835 |      632337 |      548947 |      472904 |
50:      485296 |      344273 |      289347 |      240911 |      197652 |
55:      160654 |      129093 |      101564 |      78499 |      60069 |
60:      44360 |      32554 |      23054 |      16448 |      11533 |
65:      7835 |      5295 |      3371 |      2289 |      1372 |
70:      804 |      450 |      222 |      104 |      34 |
75:      16 |      8 |      2 |      0 |      0 |
T:49995000
Running time: 0.303238

00:      2076 |      14212 |      37870 |      70863 |      113190 |
05:      162616 |      219792 |      281373 |      348231 |      418908 |
10:      492542 |      568919 |      643293 |      723787 |      799933 |
15:      878146 |      953657 |      1025635 |      1097551 |      1161963 |
20:      1225131 |      1285543 |      1340874 |      1389657 |      1435607 |
25:      1471150 |      1505326 |      1534652 |      1554099 |      1566213 |
30:      1574002 |      1572648 |      1562633 |      1558305 |      1527753 |
35:      1580146 |      1463004 |      1419940 |      1372128 |      1313777 |
40:      1249134 |      1177114 |      1097835 |      1012343 |      928271 |
45:      823626 |      723835 |      632337 |      548947 |      472904 |
50:      485296 |      344273 |      289347 |      240911 |      197652 |
55:      160654 |      129093 |      101564 |      78499 |      60069 |
60:      44360 |      32554 |      23054 |      16448 |      11533 |
65:      7835 |      5295 |      3371 |      2289 |      1372 |
70:      804 |      450 |      222 |      104 |      34 |
75:      16 |      8 |      2 |      0 |      0 |
T:49995000

Difference between CPU and GPU histograms:
00:      0 |      0 |      0 |      0 |      0 |
05:      0 |      0 |      0 |      0 |      0 |
10:      0 |      0 |      0 |      0 |      0 |
15:      0 |      0 |      0 |      0 |      0 |
20:      0 |      0 |      0 |      0 |      0 |
25:      0 |      0 |      0 |      0 |      0 |
30:      0 |      0 |      0 |      0 |      0 |

```

Project 1 for apps/GPU_course/runScript.sh proj1-feng12.cu 10000 500 256

```

srun:
srun: end of defined options
srun: jobid 228792: nodes(1):'GPU15', cpu counts: 16(x1)
srun: launch/slurm: launch_p_step_launch: CpuBindType=(null type)
srun: launching StepId=228792.0 on host GPU15, 1 tasks: 0
srun: route/default: init: route default plugin loaded
srun: launch/slurm: _task_start: Node GPU15, 1 tasks started
Running time: 1.365944
00:      2076 |      14212 |      37870 |      70863 |      113190 |
05:      162616 |      219792 |      281373 |      348231 |      418908 |
10:      492542 |      568919 |      643293 |      723787 |      799933 |
15:      878146 |      953657 |      1025635 |      1097551 |      1161963 |
20:      1225131 |      1285543 |      1340874 |      1389657 |      1435607 |
25:      1471150 |      1505326 |      1534652 |      1554099 |      1566213 |
30:      1574002 |      1572648 |      1562633 |      1558305 |      1527753 |
35:      1580146 |      1463004 |      1419940 |      1372128 |      1313777 |
40:      1249134 |      1177114 |      1097835 |      1012343 |      928271 |
45:      823626 |      723835 |      632337 |      548947 |      472904 |
50:      485296 |      344273 |      289347 |      240911 |      197652 |
55:      160654 |      129093 |      101564 |      78499 |      60069 |
60:      44360 |      32554 |      23054 |      16448 |      11533 |
65:      7835 |      5295 |      3371 |      2289 |      1372 |
70:      804 |      450 |      222 |      104 |      34 |
75:      16 |      8 |      2 |      0 |      0 |
T:49995000
Running time: 0.302135

00:      2076 |      14212 |      37870 |      70863 |      113190 |
05:      162616 |      219792 |      281373 |      348231 |      418908 |
10:      492542 |      568919 |      643293 |      723787 |      799933 |
15:      878146 |      953657 |      1025635 |      1097551 |      1161963 |
20:      1225131 |      1285543 |      1340874 |      1389657 |      1435607 |
25:      1471150 |      1505326 |      1534652 |      1554099 |      1566213 |
30:      1574002 |      1572648 |      1562633 |      1558305 |      1527753 |
35:      1580146 |      1463004 |      1419940 |      1372128 |      1313777 |
40:      1249134 |      1177114 |      1097835 |      1012343 |      928271 |
45:      823626 |      723835 |      632337 |      548947 |      472904 |
50:      485296 |      344273 |      289347 |      240911 |      197652 |
55:      160654 |      129093 |      101564 |      78499 |      60069 |
60:      44360 |      32554 |      23054 |      16448 |      11533 |
65:      7835 |      5295 |      3371 |      2289 |      1372 |
70:      804 |      450 |      222 |      104 |      34 |
75:      16 |      8 |      2 |      0 |      0 |
T:49995000

Difference between CPU and GPU histograms:
00:      0 |      0 |      0 |      0 |      0 |

```

Project 1 for apps/GPU_course/runScript.sh proj1-feng12.cu 10000 500 512

```
srun: Launching StepId=228798.0 on host GPU13, 1 tasks: 0
srun: route/default: init: route default plugin loaded
srun: launch/slurm: _task_start: Node GPU13, 1 tasks started
Running time: 1.365160

00:      2076 |      14212 |      37870 |      70863 |      113190 |
05:      162616 |      219792 |      281373 |      348231 |      418908 |
10:      492542 |      568919 |      643293 |      723787 |      799933 |
15:      878146 |      953657 |      1025635 |      1097551 |      1161963 |
20:      1225131 |      1285543 |      1340874 |      1389657 |      1435607 |
25:      1471150 |      1505326 |      1534652 |      1554499 |      1566213 |
30:      1574002 |      1572648 |      1562633 |      1550305 |      1527753 |
35:      1500146 |      1463004 |      1419940 |      1372128 |      1313777 |
40:      1249134 |      1177114 |      1097835 |      1012343 |      920271 |
45:      823626 |      723835 |      632337 |      548947 |      472904 |
50:      405296 |      344273 |      289347 |      240911 |      197652 |
55:      160654 |      129093 |      101564 |      78499 |      60069 |
60:      44360 |      32554 |      23054 |      16448 |      11533 |
65:      7835 |      5295 |      3371 |      2289 |      1372 |
70:      804 |      450 |      222 |      104 |      34 |
75:      16 |      8 |      2 |      0 |      0 |
T:49995000
Running time: 0.292884

00:      2076 |      14212 |      37870 |      70863 |      113190 |
05:      162616 |      219792 |      281373 |      348231 |      418908 |
10:      492542 |      568919 |      643293 |      723787 |      799933 |
15:      878146 |      953657 |      1025635 |      1097551 |      1161963 |
20:      1225131 |      1285543 |      1340874 |      1389657 |      1435607 |
25:      1471150 |      1505326 |      1534652 |      1554499 |      1566213 |
30:      1574002 |      1572648 |      1562633 |      1550305 |      1527753 |
35:      1500146 |      1463004 |      1419940 |      1372128 |      1313777 |
40:      1249134 |      1177114 |      1097835 |      1012343 |      920271 |
45:      823626 |      723835 |      632337 |      548947 |      472904 |
50:      405296 |      344273 |      289347 |      240911 |      197652 |
55:      160654 |      129093 |      101564 |      78499 |      60069 |
60:      44360 |      32554 |      23054 |      16448 |      11533 |
65:      7835 |      5295 |      3371 |      2289 |      1372 |
70:      804 |      450 |      222 |      104 |      34 |
75:      16 |      8 |      2 |      0 |      0 |
T:49995000

Difference between CPU and GPU histograms:

00:      0 |      0 |      0 |      0 |      0 |
05:      0 |      0 |      0 |      0 |      0 |
10:      0 |      0 |      0 |      0 |      0 |
15:      0 |      0 |      0 |      0 |      0 |
```

SO I compare the execution times of Project 1 (CPU-based SDH computation) and Project 2 (GPU-accelerated SDH using CUDA) under different block sizes. The results are presented in milliseconds for consistency and better comparison.

For Project 1 (CPU execution), the processing time for 10,000 samples with a bucket width of 500 was measured for different block sizes:

Block size 128: 303.238 ms

Block size 256: 302.135 ms

Block size 512: 292.884 ms

The CPU execution time slightly improves as the block size increases, likely due to better memory access patterns and loop optimizations.

For Project 2 (GPU execution using CUDA), the same dataset and bucket width were tested with varying block sizes, and the recorded execution times were:

Block size 128: 21.610 ms

Block size 256: 25.676 ms

Block size 512: 31.379 ms

The results indicate that GPU acceleration significantly reduces computation time, with the best performance observed at block size 128, achieving 21.610 ms. Increasing the block size to 256 and 512 resulted in slower execution, with the worst performance at block size 512 (31.379 ms). This slowdown is attributed to higher shared memory usage, increased atomic operation contention, and suboptimal thread scheduling.

Overall, GPU execution is approximately 14 times faster than CPU execution, demonstrating the efficiency of parallel processing. The optimal block size for CUDA in this case is 128, as it provides the best balance between occupancy, memory efficiency, and computational throughput. Increasing the block size beyond this value leads to performance degradation due to resource contention.